

Machine Learning Basics

Machine Learning is a subset of Artificial Intelligence that enables computers to learn from data without being explicitly programmed.

Types of Machine Learning:

1. Supervised Learning:

The model learns from labeled data.

Examples:

- Linear Regression
- Logistic Regression
- Decision Trees
- Support Vector Machines

2. Unsupervised Learning:

The model learns patterns from unlabeled data.

Examples:

- K-Means Clustering
- Hierarchical Clustering
- DBSCAN

3. Reinforcement Learning:

The agent learns by interacting with an environment and receiving rewards or penalties.

Common Steps in Machine Learning:

- Data Collection
- Data Cleaning
- Feature Engineering
- Model Training
- Model Evaluation
- Deployment

Evaluation Metrics:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC